TOPIC 2 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

First-Order Equations: Qualitative and Analytical Methods

Two complementary ways of understanding a first-order equation: the qualitative phase line, which shows how every solution behaves without solving anything, and the analytical methods (separation, integrating factor, exact equations) that produce explicit formulas. Each method is derived, justified and applied to a physical model.

Ground truth: Robinson (2004) Ch. 7 (pp. 46–58), Ch. 8 (pp. 59–74), Ch. 9 (pp. 75–88), Ch. 10 (pp. 89–98) • Lecture map ODE_lecture_theory_to_Robinson_chapters.txt , Part I (Sep 10, Sep 17)

≈ 30 min study time • 35 displayed equations • 6 figures

SECTION 01 • ROBINSON SEC. 7.1 • LECTURE SEP 10

1. Autonomous equations and the phase line

For a general first-order equation we will look for explicit formulas. For the special class of **autonomous** equations

$$\frac{dx}{dt} = f(x) \tag{7.1}$$

there is a faster and completely rigorous alternative: describe *how* the solutions behave ("every solution starting between 0 and 1 tends to 1") without writing them down. The method only needs the sign of f [Robinson, Ch. 7 intro, p. 46].

1.1 The particle picture

By Theorem 6.2 (Topic 1), if f and f' are continuous, a solution of (7.1) is completely determined by its value at any one time. Think of $x(t)$ as the position of a particle moving on a line, with velocity $\dot{x} = f(x)$ that depends only on where the particle is [Sec. 7.1, p. 46].

DEFINITION — STATIONARY POINT [ROBINSON, P. 47]

A **stationary point** of (7.1) is a value x^* with $f(x^*) = 0$. The constant function $x(t) \equiv x^*$ is then a solution, since $\dot{x} = 0 = f(x^*)$. (Other names: equilibrium point, critical point, fixed point; Robinson reserves the last two for other meanings.)

- Where $f(x) > 0, \dot{x} > 0$: $x(t)$ increases and the particle moves **right**.

- Where $f(x) < 0, \dot{x} < 0$: $x(t)$ decreases and the particle moves **left**.

PROPOSITION — SOLUTIONS ARE MONOTONE AND NEVER REACH A STATIONARY POINT IN FINITE TIME

If f and f' are continuous, a non-constant solution of $\dot{x} = f(x)$ cannot change direction and cannot cross or reach a stationary point.

PROOF [ROBINSON P. 47; EXERCISE 7.8]

Suppose the solution reversed direction at some time t^*. Then it is instantaneously at rest, $\dot{x}(t^*) = f(x(t^*)) = 0$, so $x^* = x(t^*)$ is a stationary point. But $x(t) \equiv x^*$ is also a solution through the same point at the same time, and by uniqueness the two solutions coincide: the particle never moved, a contradiction. The same argument rules out reaching a stationary point at any finite time. So $\dot{x} = f(x(t))$ never vanishes, and by continuity it keeps one sign: $x(t)$ is strictly monotone.

■

1.2 Drawing the phase diagram

1. Sketch the graph of $f(x)$ against x (only its sign matters).
2. Mark the zeros x^* of f on a horizontal line: these are the stationary points.
3. On each interval between stationary points draw an arrow to the right if $f > 0$ there and to the left if $f < 0$.
4. Read off the behaviour: a solution moves in the direction of the arrow and, being monotone, either tends to the next stationary point or to $\pm\infty$ (Robinson Exercise 7.10).

The resulting picture is the **phase diagram** (or **phase portrait**) [p. 47, Figs. 7.1-7.2].

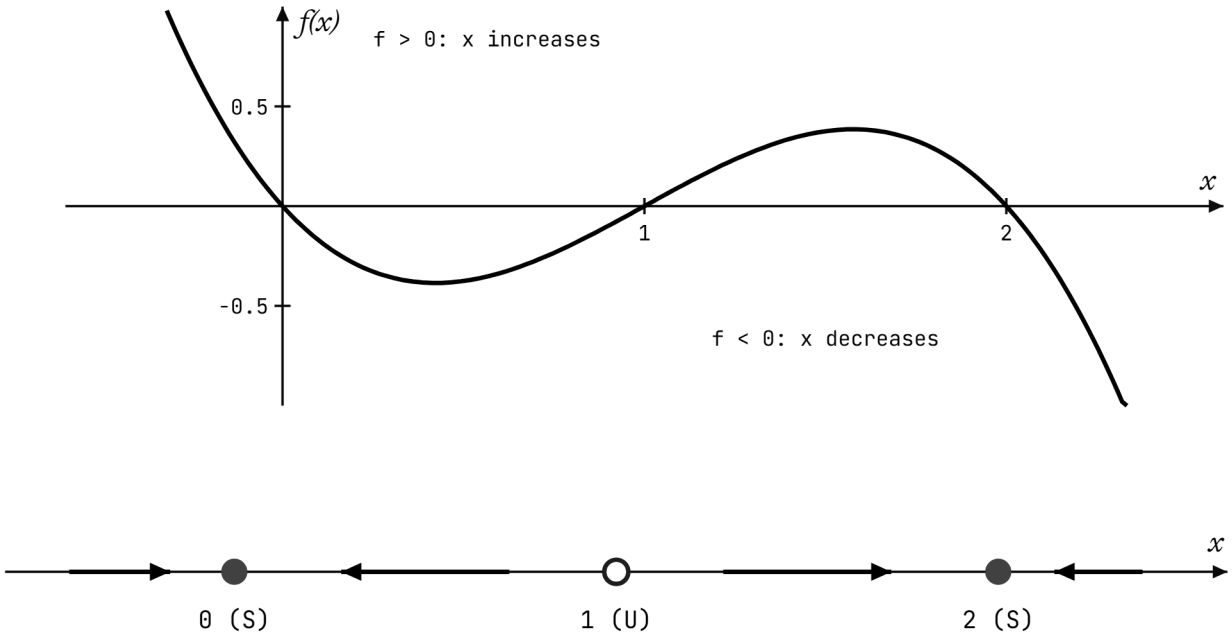


Figure 2.1. *Building a phase diagram (method of Robinson Figs. 7.1–7.2) for $\dot{x} = -x(1-x)(2-x)$, Exercise 7.1 (iv). Top: the graph of f. Bottom: the phase line. Arrows point right where $f > 0$ and left where $f < 0$; filled points are stable (S), hollow points unstable (U).*

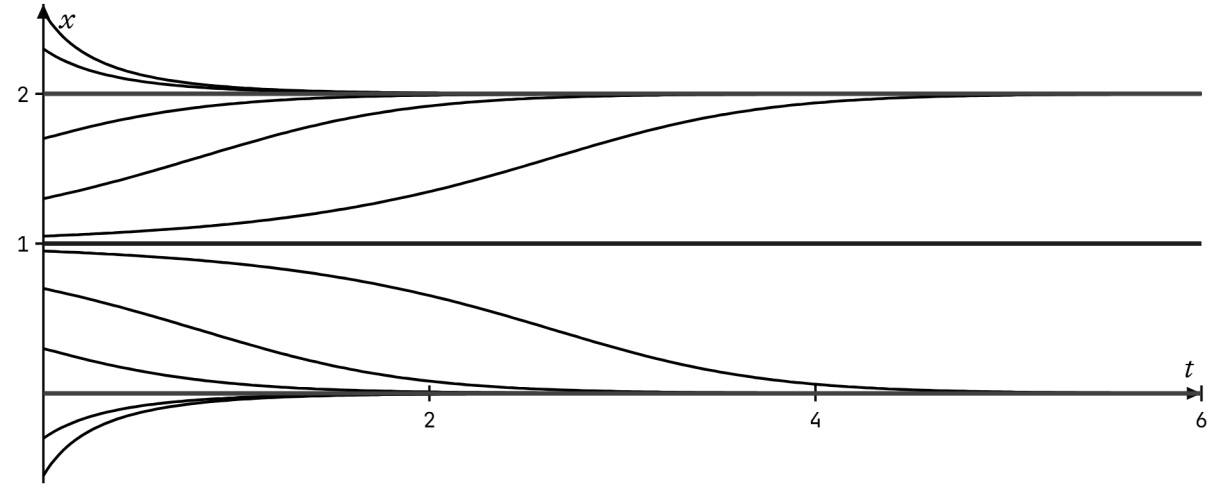


Figure 2.2. *Solutions $x(t)$ of the same equation from several initial values (numerical integration). They are monotone, never cross the constant solutions $x = 0, 1, 2$ and approach the stable ones; compare Robinson Fig. 7.8.*

2. Stability of stationary points

DEFINITIONS — STABLE, ATTRACTING, UNSTABLE [ROBINSON, SEC. 7.2, EQS. (7.2)–(7.3), P. 48]

- x^* is **stable** if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x_0 - x^*| < \delta \implies |x(t) - x^*| < \varepsilon \quad \text{for all } t \geq 0. \tag{7.2}$$

  "Start close enough and you stay close."
- x^* is **attracting** if there is a $\delta > 0$ such that

$$|x_0 - x^*| < \delta \implies x(t) \to x^* \text{ as } t \to +\infty. \tag{7.3}$$

- x^* is **unstable** if it is not stable: there is an $\varepsilon > 0$ such that for every $\delta > 0$ some x_0 with $|x_0 - x^*| < \delta$ gives $|x(t) - x^*| > \varepsilon$ for some $t > 0$.

In one dimension attracting implies stable (Exercise 7.9), but stable does not imply attracting. Robinson's counterexample is $\dot{x} = -x$ for $x < 0$, $\dot{x} = 0$ for $0 \leq x \leq 1$, $\dot{x} = 1 - x$ for $x > 1$: every point of $[0, 1]$ is stationary and stable, but none is attracting, because nearby points in $[0, 1]$ do not move at all [p. 48; Fig. 7.3, p. 49].

2.1 The derivative test

STABILITY CRITERION [ROBINSON, SEC. 7.3, P. 49, FIG. 7.5]

Let $f(x^*) = 0$ with f continuously differentiable.

- If $f'(x^*) < 0$, then x^* is **stable** (in fact attracting).
- If $f'(x^*) > 0$, then x^* is **unstable**.
- If $f'(x^*) = 0$, the test is inconclusive: the point may be stable, unstable, or **semi-stable** (stable on one side, unstable on the other) [Fig. 7.6, p. 50].

Why it works (graphical argument, as in the book). If $f'(x^*) < 0$, the graph of f crosses zero going downwards: $f > 0$ just to the left of x^* and $f < 0$ just to the right. Both arrows point towards x^*. If $f'(x^*) > 0$ the signs are reversed and both arrows point away.

Why it works (linearisation – complementary derivation). Write $x(t) = x^* + \xi(t)$ with a small perturbation ξ. Since x^* is constant, $\dot{x} = \dot{\xi}$. Taylor's theorem gives $f(x^* + \xi) = f(x^*) + f'(x^*)\,\xi + O(\xi^2) = f'(x^*)\,\xi + O(\xi^2)$. Dropping the quadratic term,

$$\dot{\xi} \approx f'(x^*)\,\xi \quad \Longrightarrow \quad \xi(t) \approx \xi(0)\,e^{f'(x^*)\,t}.$$

The perturbation decays if $f'(x^*) < 0$ and grows if $f'(x^*) > 0$. The quantity $1/|f'(x^*)|$ is the time scale of the approach or departure. When $f'(x^*) = 0$ the linear term vanishes and the neglected $O(\xi^2)$ terms decide, which is why the test is then silent.

EXAMPLE — THE CUBIC OF FIGURE 2.1

$f(x) = -x(1-x)(2-x) = -(x^3 - 3x^2 + 2x)$, so $f'(x) = -(3x^2 - 6x + 2)$. Then $f'(0) = -2 < 0$ (stable), $f'(1) = -(3 - 6 + 2) = 1 > 0$ (unstable), $f'(2) = -(12 - 12 + 2) = -2 < 0$ (stable), in agreement with the arrows.

EXAMPLE — LOGISTIC POPULATION (ROBINSON SEC. 7.5.1, EQ. (7.4), P. 51)

$\dot{p} = f(p) = kp(1 - p/M)$, with $f'(p) = k - 2kp/M$ (product rule on $kp \cdot (1 - p/M)$, or expand $kp - kp^2/M$ first). Stationary points $p = 0$ and $p = M$: $f'(0) = k > 0$, so extinction is **unstable**; $f'(M) = k - 2k = -k < 0$, so the carrying capacity is **stable**. Every positive population tends to M, with relaxation time $1/k$ near M. The phase diagram cannot, however, give k or M from data; that needs the explicit solution of Topic 1, Section 5.4.

EXAMPLE — TERMINAL VELOCITY (ROBINSON SEC. 7.5.2, EQ. (7.5), P. 52)

For downward speed $v > 0$: $\dot{v} = f(v) = g - (k/m)v^2$. The unique positive stationary point is $v^* = \sqrt{mg/k}$, and $f'(v) = -2kv/m$, so $f'(v^*) = -2\sqrt{gk/m} < 0$: terminal velocity is stable and attracts every initial speed. The explicit $\tanh$ solution is derived in Topic 1, Section 6.5.

WHAT THE PHASE LINE CANNOT TELL YOU (ROBINSON SEC. 7.5.3, P. 53)

$\dot{x} = |x|$ and $\dot{x} = x^2$ have the same phase diagram (one semi-stable point at 0, arrows pointing right everywhere). But for $x_0 > 0$ the first gives $x = x_0 e^t$, defined for all time, while the second gives $x = 1/(x_0^{-1} - t)$, which blows up at $t = 1/x_0$. The phase line captures the direction of motion, not its speed.

SECTION 03 • ROBINSON SEC. 7.4, 7.6, EX. 7.5-7.6

3. Structural stability and bifurcations

Models contain parameters (a growth rate, a load, a drag coefficient) that are never known exactly. It matters whether a small change in a parameter changes the picture qualitatively.

DEFINITIONS [ROBINSON, SEC. 7.4, P. 50]

If a small change to f changes the phase diagram drastically, the equation has undergone a **bifurcation**. If no small change to f can change the qualitative phase diagram, $\dot{x} = f(x)$ is **structurally stable**.

If $f'(x^*) \neq 0$, a small change to f (for instance adding a small constant, which shifts the graph up or down) still leaves a nearby zero with the same sign of slope, hence a nearby stationary point of the same type (Exercise 7.11). So **a bifurcation near x^* requires $f'(x^*) = 0$** – exactly the degenerate case of the derivative test.

3.1 The pitchfork bifurcation

Robinson's worked example `[Sec. 7.6, Eq. (7.6), pp. 54–56]`:

$$\dot{x} = x\,(k - x^2), \qquad f(x) = kx - x^3, \qquad f'(x) = k - 3x^2.$$

- $k < 0$**:** $k - x^2 < 0$ for every x, so the only stationary point is $x^* = 0$, with $f'(0) = k < 0$: stable.
- $k = 0$**:** $f(x) = -x^3$ and $f'(0) = 0$ – the test fails. The sign of f (> 0 for $x < 0$, < 0 for $x > 0$) shows 0 is still stable. The vanishing derivative signals a possible bifurcation.
- $k > 0$**:** three stationary points $x^* = 0, \pm\sqrt{k}$. $f'(0) = k > 0$: unstable. $f'(\pm\sqrt{k}) = k - 3k = -2k < 0$: both stable.

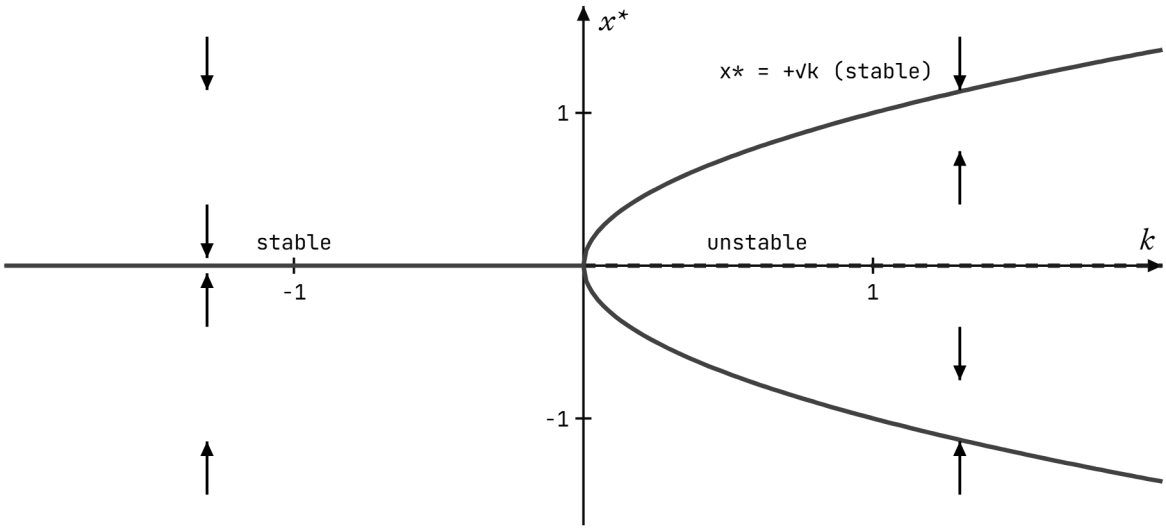


Figure 2.3. *Bifurcation diagram of the pitchfork $\dot{x} = x(k - x^2)$ (Robinson Sec. 7.6, Fig. 7.14). Solid green: stable stationary points; dashed red: unstable. At $k = 0$ the single stable point $x^* = 0$ splits into three.*

PHYSICAL READING (COMPLEMENTARY)

The pitchfork is the generic picture of **symmetric buckling**: a slender strut under axial load stays straight ($x^* = 0$) below the critical load and, above it, the straight state becomes unstable and the strut deflects to one of two mirror-image positions $\pm\sqrt{k}$. Which one is chosen depends on imperfections – the same sensitivity that makes $k = 0$ structurally unstable.

3.2 Saddle-node and transcritical bifurcations (Exercises 7.5 and 7.6)

- **Saddle-node:** $\dot{x} = x^2 - k$. For $k < 0$ there are no stationary points; for $k > 0$ there are two, $x^* = \pm\sqrt{k}$, with $f'(x) = 2x$: $-\sqrt{k}$ stable ($f' = -2\sqrt{k}$), $+\sqrt{k}$ unstable. They are born together at $k = 0$.
- **Transcritical:** $\dot{x} = kx - x^2 = x(k - x)$, stationary points 0 and k, $f'(x) = k - 2x$. Then $f'(0) = k$ and $f'(k) = -k$: for $k < 0$, 0 is stable and k unstable; for $k > 0$ the roles swap. The two points exchange stability as they cross.

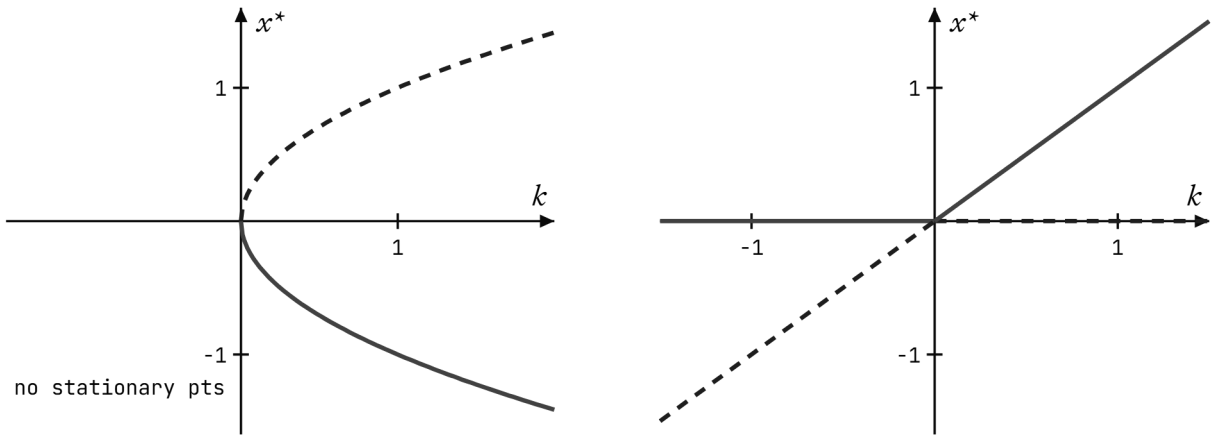


Figure 2.4. *Left: saddle-node bifurcation of $\dot{x} = x^2 - k$ (Robinson Exercise 7.5): two stationary points collide and disappear as k decreases through 0. Right: transcritical bifurcation of $\dot{x} = kx - x^2$ (Exercise 7.6): the branches $x^* = 0$ and $x^* = k$ cross and exchange stability. Green solid: stable; red dashed: unstable.*

SECTION 04 • ROBINSON SEC. 8.1, 8.4, 8.6 • LECTURE SEP 10

4. Separable equations

DEFINITION — SEPARABLE EQUATION [ROBINSON, EQ. (8.1), P. 59]

$$\frac{dx}{dt} = f(x)\, g(t). \tag{8.1}$$

The right-hand side is a product of a function of x only and a function of t only. Autonomous equations ($g \equiv 1$) and "trivial" equations $\dot{x} = g(t)$ ($f \equiv 1$) are special cases.

4.1 The recipe

Divide by $f(x)$, "multiply by dt" and integrate both sides [Sec. 8.1, Eqs. (8.2)-(8.3), p. 60]:

$$\int \frac{dx}{f(x)} = \int g(t)\,dt, \qquad \text{or, with } x(t_0)=x_0: \qquad \int_{x_0}^{x(t)} \frac{dx}{f(x)} = \int_{t_0}^{t} g(\tilde{t})\,d\tilde{t}. \tag{8.3}$$

On the left the limits are the values of x at the times t_0 and t; on the right they are the times themselves. Robinson warns that "multiplying by dt" should make you uneasy – the next subsection shows why it nevertheless gives the right answer.

4.2 Justification [Robinson, Sec. 8.4, pp. 64-65]

1. **Constant solutions first.** If $f(x(s))=0$ at some time s, then $x(t)\equiv x(s)$ solves the equation ($\dot{x}=0=f(x(s))g(t)$), and by uniqueness it is the solution. So either $f(x(t))=0$ for every t, or $f(x(t))\neq 0$ for every t. These constant solutions are lost when dividing by f and must be listed separately.

2. In the second case divide: $\dfrac{1}{f(x)}\dfrac{dx}{dt} = g(t)$ (8.8).

3. Let H be an antiderivative of $1/f$, $H'(x)=1/f(x)$. By the chain rule

$$\frac{d}{dt}H\big(x(t)\big) = H'\big(x(t)\big)\,\frac{dx}{dt} = \frac{1}{f(x)}\frac{dx}{dt} = g(t).$$

4. Let G be an antiderivative of g. Two functions with the same derivative differ by a constant: $H(x(t)) = G(t) + c$ (8.9). At $t=t_0$: $c = H(x_0) - G(t_0)$, so $H(x(t)) - H(x_0) = G(t) - G(t_0)$, which by Barrow's rule is exactly (8.3) = (8.11).

No infinitesimals were manipulated: the "recipe" is the chain rule read backwards. To obtain $x(t)$ explicitly one must invert H; sometimes the implicit relation $H(x)=G(t)+c$ is the best available.

EXAMPLE 8.1 (ROBINSON P. 60): $\dot{x}=x^2,\ x(0)=x_0$

If $x_0=0$, $x\equiv 0$. Otherwise separate, $x^{-2}dx = dt$, and integrate between corresponding limits. An antiderivative of x^{-2} is $-x^{-1}$:

$$\int_{x_0}^{x(t)} \frac{dx}{x^2} = \int_0^t d\tilde{t} \implies \Big[-\frac{1}{x}\Big]_{x_0}^{x(t)} = t \implies -\frac{1}{x(t)} + \frac{1}{x_0} = t \implies x(t) = \frac{1}{x_0^{-1} - t}.$$

This is the blow-up solution (6.7) of Topic 1, now derived rather than guessed.

REMOVING MODULUS SIGNS WITH THE PHASE DIAGRAM

Integrals of $1/x$ produce $\ln|x|$. Robinson's advice is to remove the modulus with the phase diagram: a solution cannot cross a stationary point, so $x(t)$ keeps the sign of x_0 (for $\dot{x}=\lambda x$) or the sign of $M-p_0$ (for the logistic equation). Both full derivations are in Topic 1, Section 5.

4.3 Partial fractions: the pitchfork equation solved [Robinson, Sec. 8.6.1, pp. 68–70]

Solve $\dot{x} = x(\kappa^2 - x^2)$, $x(0) = x_0$ (the pitchfork with $k = \kappa^2 > 0$). Separating,

$$\frac{dx}{x(\kappa^2 - x^2)} = dt, \qquad \frac{1}{x(\kappa - x)(\kappa + x)} = \frac{1}{\kappa^2}\left[\frac{1}{x} + \frac{1}{2(\kappa - x)} - \frac{1}{2(\kappa + x)}\right].$$

(Check by putting the bracket over the common denominator $2x(\kappa - x)(\kappa + x)$: the numerator is $2(\kappa^2 - x^2) + x(\kappa + x) - x(\kappa - x) = 2\kappa^2 - 2x^2 + 2x^2 = 2\kappa^2$.) Integrate from 0 to t; with $\int \frac{dx}{\kappa - x} = -\ln|\kappa - x|$ (substitution $u = \kappa - x$):

$$\Big[\ln|x| - \tfrac{1}{2}\ln|\kappa - x| - \tfrac{1}{2}\ln|\kappa + x|\Big]_{x_0}^{x(t)} = \kappa^2 t \implies \ln\frac{|x(t)|\sqrt{|\kappa^2 - x_0^2|}}{|x_0|\sqrt{|\kappa^2 - x(t)^2|}} = \kappa^2 t.$$

Exponentiate and square: $x(t)^2|\kappa^2 - x_0^2| = x_0^2|\kappa^2 - x(t)^2|e^{2\kappa^2 t}$ (8.15). The phase diagram shows that $x^2 - \kappa^2$ keeps its sign, so the moduli go, and solving the linear equation for $x(t)^2$:

$$x(t) = \pm\sqrt{\frac{\kappa^2}{1 + e^{-2\kappa^2 t}\left(\kappa^2 x_0^{-2} - 1\right)}}, \tag{8.16}$$

with the sign of x_0. As $t \to \infty$, $e^{-2\kappa^2 t} \to 0$ and $x \to \pm\kappa$: the two stable branches of the pitchfork.

SECTION 05 • ROBINSON CH. 9 • LECTURE SEP 17

5. First-order linear equations: the integrating factor

The general linear first-order equation is $a_1(t)\dot{x} + a_0(t)x = f(t)$. Where $a_1 \neq 0$ we divide by it and obtain the standard form [Robinson, Eq. (9.1), p. 75]

$$\frac{dx}{dt} + p(t)\,x = q(t). \tag{9.1}$$

It is separable when p and q are constants (and in a few special cases such as $q \equiv 0$); the integrating factor works for all p, q.

5.1 The idea, with constant coefficients [Sec. 9.1, pp. 75–76]

For $\dot{x} + px = q$ with constants p, q, notice that by the product rule

$$\frac{d}{dt}\big(x\,e^{pt}\big) = \dot{x}\,e^{pt} + x\,p\,e^{pt} = e^{pt}\big(\dot{x} + px\big). \tag{9.3}$$

So multiplying the equation by e^{pt} turns its left side into an exact derivative:

$$\frac{d}{dt}\big(xe^{pt}\big) = qe^{pt} \implies xe^{pt} = \frac{q}{p}e^{pt} + C \implies x(t) = \frac{q}{p} + Ce^{-pt}. \tag{9.5}$$

If $p > 0$, every solution tends to q/p as $t \to \infty$, whatever the initial condition. With $x(a) = x_a$: $C = (x_a - q/p)e^{pa}$ and $x(t) = q/p + (x_a - q/p)e^{-p(t-a)}$ (9.6).

5.2 Derivation of the integrating factor [Sec. 9.2, pp. 76–78]

For $\dot{x} + p(t)x = q(t)$ look for a function $I(t) > 0$ such that, after multiplying by I, the left side $I\dot{x} + Ipx$ is the derivative of Ix.

1. Product rule: $\dfrac{d}{dt}(Ix) = \dot{I}\,x + I\,\dot{x}$.
2. We need $\dot{I}\,x + I\dot{x} = I\dot{x} + Ipx$, i.e. $\dot{I}\,x = Ipx$ for all x, i.e. $\dfrac{dI}{dt} = p(t)\,I$.
3. This is separable: $\dfrac{dI}{I} = p(t)\,dt$, so $\ln|I| = \int p(t)\,dt$. Choosing the positive root and dropping the constant (any non-zero multiple of an integrating factor is still one):

INTEGRATING FACTOR AND SOLUTION FORMULA

$$I(t) = \exp\left(\int p(t)\,dt\right) = e^{P(t)}, \qquad P'(t) = p(t).$$

$$\frac{d}{dt}\Big[x(t)\,e^{P(t)}\Big] = q(t)\,e^{P(t)} \qquad \text{(Robinson p. 78; (9.8) is the equation itself)}$$

Integrating from t_0 to t and using Barrow's rule on the left, $\big[x\,e^{P}\big]_{t_0}^{t} = x(t)e^{P(t)} - x_0e^{P(t_0)}$, so

$$x(t) = e^{-P(t)}\left[x_0\,e^{P(t_0)} + \int_{t_0}^{t} q(s)\,e^{P(s)}\,ds\right].$$

The two terms have a clear meaning: the first is the free response, the initial value carried forward by the homogeneous dynamics $e^{-(P(t)-P(t_0))}$; the second is the forced response, the accumulated effect of the input q. This "homogeneous + particular" structure reappears for second-order equations in Topic 3.

EXAMPLE 9.1 (ROBINSON PP. 78–79): $\dot{x} + 3x = t,\ x(0) = 8/9$

1. $p = 3$, so $I = e^{\int 3\,dt} = e^{3t}$ and $\dfrac{d}{dt}\big(xe^{3t}\big) = te^{3t}$.

2. Antiderivative of se^{3s} by parts, with $u = s, dv = e^{3s}ds, du = ds, v = e^{3s}/3$: $\displaystyle\int se^{3s}ds = \frac{se^{3s}}{3} - \int \frac{e^{3s}}{3}ds = \frac{se^{3s}}{3} - \frac{e^{3s}}{9}$.

3. Integrate from 0 to t (Barrow):

$$x(t)e^{3t} - x(0) = \Big[\frac{se^{3s}}{3} - \frac{e^{3s}}{9}\Big]_0^t = \frac{te^{3t}}{3} - \frac{e^{3t}}{9} - \Big(0 - \frac{1}{9}\Big).$$

4. With $x(0) = 8/9$: $xe^{3t} = \dfrac{te^{3t}}{3} - \dfrac{e^{3t}}{9} + 1$, so $x(t) = e^{-3t} + \dfrac{t}{3} - \dfrac{1}{9}$.

Check: $\dot{x} + 3x = (-3e^{-3t} + \tfrac{1}{3}) + (3e^{-3t} + t - \tfrac{1}{3}) = t$; $x(0) = 1 - \tfrac{1}{9} = \tfrac{8}{9}$.

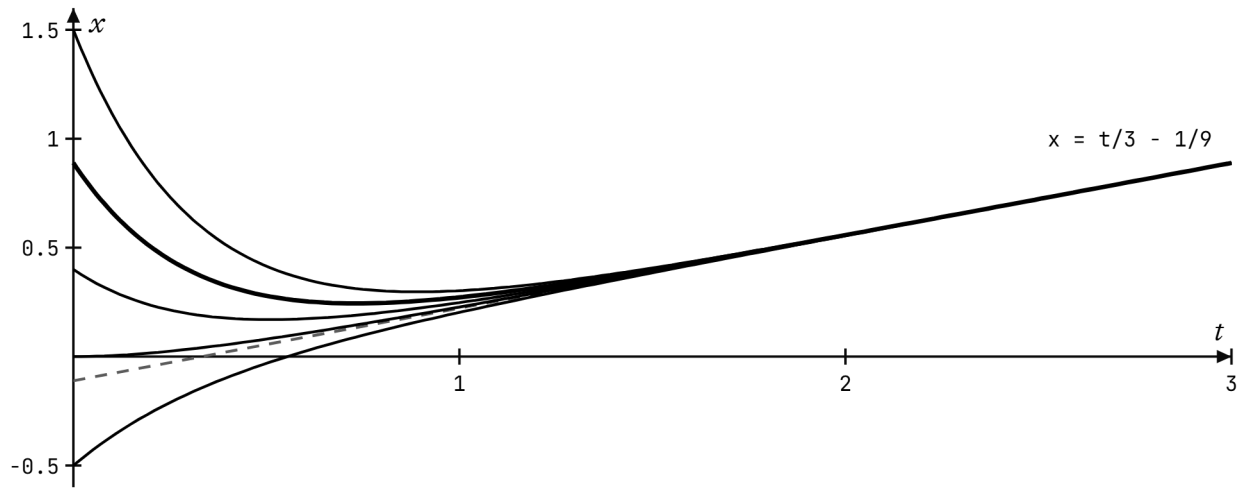


Figure 2.5. *Solutions of $\dot{x} + 3x = t$ (Robinson Example 9.1) for several $x(0)$; the thick curve is $x(0) = 8/9$. Every solution approaches the line $x = t/3 - 1/9$ because the term Ce^{-3t} decays.*

EXAMPLE 9.2 (ROBINSON P. 79): $(x^2+1)\,y' + 4xy = 12x$

1. Standard form (divide by $x^2 + 1 > 0$): $y' + \dfrac{4x}{x^2+1}y = \dfrac{12x}{x^2+1}$.

2. Integrating factor: substitute $u = x^2 + 1, du = 2x\,dx$: $\displaystyle\int \frac{4x}{x^2+1}dx = \int \frac{2\,du}{u} = 2\ln u$, so $I = e^{2\ln(x^2+1)} = (x^2+1)^2$.

3. $\dfrac{d}{dx}\big[y(x^2+1)^2\big] = 12x(x^2+1)$. With the same substitution, $\int 12x(x^2+1)\,dx = \int 6u\,du = 3u^2 = 3(x^2+1)^2$.

(4) $y(x^2+1)^2 = 3(x^2+1)^2 + c$, so $y(x) = 3 + \dfrac{c}{(x^2+1)^2}$.

5.3 Newton's law of cooling [Robinson, Sec. 9.4, Eq. (9.9), pp. 79–86]

An object at temperature $T(t)$ in surroundings at temperature $A(t)$ exchanges heat at a rate proportional to the temperature difference:

$$\frac{dT}{dt} = -k\big(T - A(t)\big), \qquad k > 0 \ [\mathrm{s}^{-1}]. \tag{9.9}$$

In standard form, $\dot T + kT = kA(t)$, with integrating factor e^{kt}.

CONSTANT SURROUNDINGS: ESTIMATING THE TIME OF DEATH (SEC. 9.4.1, PP. 80–81)

With A constant, integrating $\frac{d}{dt}[Te^{kt}] = kAe^{kt}$ between t_1 and t_2 gives $T(t_2)e^{kt_2} - T(t_1)e^{kt_1} = A\big(e^{kt_2} - e^{kt_1}\big)$, i.e.

$$T(t_2) = A + \big[T(t_1) - A\big]e^{-k(t_2 - t_1)}.$$

Data (time in hours): room at $A = 24\,^\circ\mathrm{C}$, body at $28\,^\circ\mathrm{C}$ at 8 a.m. and $26\,^\circ\mathrm{C}$ at 9 a.m. Then $26 = 24 + 4e^{-k}$, so $e^{-k} = 1/2$ and $k = \ln 2\ \mathrm{h}^{-1}$. The time of death t_0 satisfies $T(t_0) = 37$: $28 = 24 + 13e^{-k(8 - t_0)}$, so $4 = 13e^{-k(8-t_0)}$ and

$$8 - t_0 = \frac{\ln 4 - \ln 13}{-\ln 2} = \frac{\ln(13/4)}{\ln 2} = \frac{1.1787}{0.6931} = 1.70\ \mathrm{h}.$$

ERRATUM — ROBINSON P. 81

The book's equation for $8 - t_0$ is correct, but it then states "$t_0 \approx 1.7$, putting the time of death at approximately 1:42 a.m.". The value 1.7 is $8 - t_0$, not t_0. The correct result is $t_0 = 8 - 1.70 = 6.30$, i.e. about **6:18 a.m.**

Check by substitution. With $t_0 = 6.30$: $T(8) = 24 + 13 \cdot 2^{-1.70} = 24 + 13 \times 0.308 = 28.0\,^\circ\mathrm{C}$. With the book's $t_0 = 1.7$: $T(8) = 24 + 13 \cdot 2^{-6.3} = 24 + 13 \times 0.0127 = 24.2\,^\circ\mathrm{C} \neq 28$. Physically, a body that cools by $2\,^\circ\mathrm{C}$ per hour near $28\,^\circ\mathrm{C}$ cannot have been at $37\,^\circ\mathrm{C}$ more than six hours earlier.

Periodic surroundings: amplitude attenuation and lag (Sec. 9.4.2–9.4.4)

Take $A(t) = \mu + a\cos\omega t$ (mean μ, amplitude a, period $2\pi/\omega$). Integrating (9.9) from 0 to t gives $T(t)e^{kt} - T(0) = k\int_0^t A(\tilde t)e^{k\tilde t}\,d\tilde t$ (9.10). Using the antiderivative (9.11) $\int e^{kt}\cos\omega t\,dt = \dfrac{e^{kt}(k\cos\omega t + \omega\sin\omega t)}{k^2+\omega^2}$ (verify by differentiating) and combining $A\cos\omega t + B\sin\omega t = \sqrt{A^2+B^2}\cos(\omega t - \phi)$, $\tan\phi = B/A$ (9.12)–(9.15), Robinson obtains

$$T(t) = \mu + \left(T(0) - \mu - \frac{ak^2}{k^2+\omega^2} \right) e^{-kt} + \frac{ak}{\sqrt{k^2+\omega^2}} \cos(\omega t - \phi), \qquad \phi = \tan^{-1}\frac{\omega}{k}. \tag{9.17}$$

- **Transient:** the middle term decays on the time scale $1/k$ and the memory of $T(0)$ is lost.
- **Attenuation:** the inside oscillation has amplitude $a\,k/\sqrt{k^2+\omega^2} < a$ (9.18).
- **Lag:** it peaks a time $\phi/\omega = \omega^{-1}\tan^{-1}(\omega/k)$ after the outside (9.19).
- **Limits:** $k \gg \omega$ (good thermal contact): amplitude $\to a$, lag $\to 0$, the object follows its surroundings. $k \ll \omega$ (heavy insulation): amplitude $\approx ak/\omega \to 0$, lag $\to \pi/(2\omega)$, a quarter period.

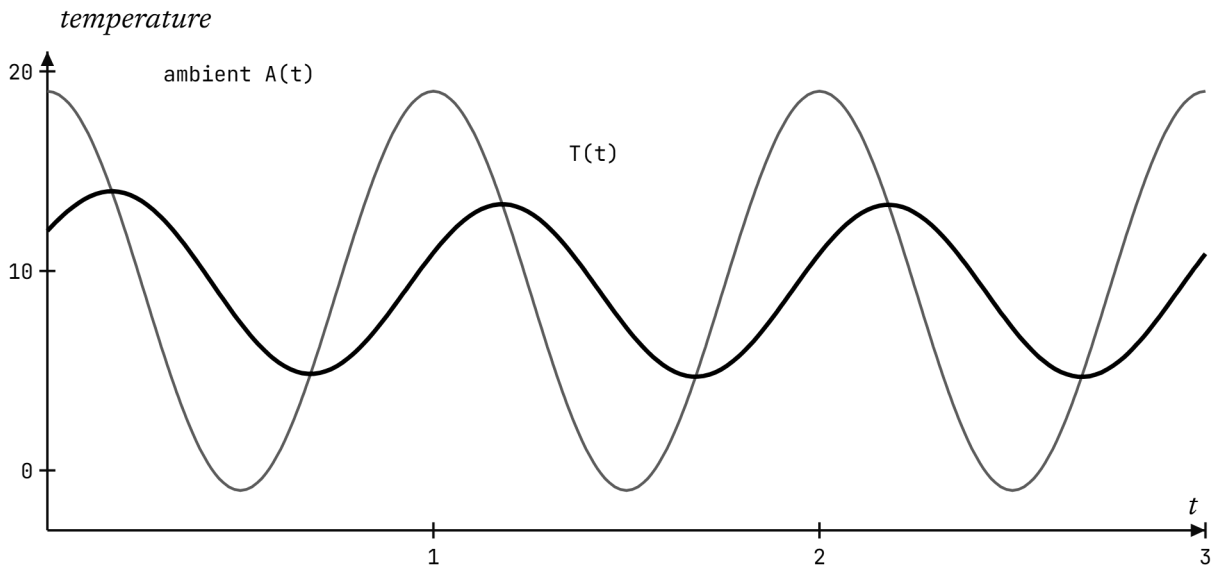


Figure 2.6. *Newton's law of cooling with a periodic surrounding temperature $A(t) = 9 + 10\cos 2\pi t$, $k = 3$, $T(0) = 12$ (data of Robinson Fig. 9.3). After the transient dies out, T oscillates with the same period, a smaller amplitude and a time lag.*

AEROSPACE MODEL — CABIN VENTILATION AS A STIRRED TANK (COMPLEMENTARY, BEYOND THE REFERENCE TEXT)

A well-mixed volume V [m³] (an aircraft cabin, a combustor, a continuously stirred tank) receives a volumetric flow Q [m³/s] of air with contaminant concentration c_{in}, and the same flow leaves with the cabin concentration $c(t)$ (perfect mixing). Occupants add a source S [kg/s]. The mass balance "rate of change = in − out + source" reads

$$\frac{d}{dt}(Vc) = Qc_{\text{in}} - Qc + S \quad\Longrightarrow\quad \frac{dc}{dt} + \frac{Q}{V}\,c = \frac{Qc_{\text{in}} + S}{V}.$$

This is (9.1) with constant $p = Q/V$ and $q = (Qc_{\text{in}} + S)/V$, so by (9.6) $c(t) = c_\infty + (c_0 - c_\infty)e^{-t/\tau}$, with steady state $c_\infty = c_{\text{in}} + S/Q$ and **residence time** $\tau = V/Q$. For a cabin of $V = 150\ \text{m}^3$ supplied with $Q = 0.5\ \text{m}^3/\text{s}$, $\tau = 300$ s: after a disturbance the air is 95% refreshed in $3\tau = 15$ minutes ($e^{-3} \approx 0.05$). Dimensions: $[Q/V] = \text{s}^{-1}$, $[S/Q] = \text{kg/m}^3$. Newton's law of cooling is the same equation with $k = Q/V$.

6. Exact equations

Start from the answer. A relation $F(x, y) = c$ defines curves $y(x)$ implicitly [Eq. (10.1), p. 89]. Differentiating $F(x, y(x)) = c$ with respect to x by the multivariable chain rule gives a first-order ODE:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0. \tag{10.2}$$

Exact equations are the ODEs that arise this way; solving them means recovering F.

DEFINITION — EXACT EQUATION [ROBINSON, EQS. (10.3)-(10.4), P. 89]

The equation

$$f(x, y) + g(x, y)\frac{dy}{dx} = 0 \tag{10.3}$$

is **exact** if there is a function $F(x, y)$ (a potential) with $\dfrac{\partial F}{\partial x} = f$ and $\dfrac{\partial F}{\partial y} = g$. Its solutions are then given implicitly by $F(x, y) = c$.

TEST FOR EXACTNESS [ROBINSON, EQ. (10.5), P. 90]

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x} \tag{10.5}$$

is necessary and sufficient for (10.3) to be exact (for smooth f, g on a rectangle).

PROOF

Necessity. If $f = F_x$ and $g = F_y$, then $f_y = F_{xy}$ and $g_x = F_{yx}$, and mixed second partial derivatives of a smooth function are equal.

Sufficiency (constructive). Define F by integrating f in x, holding y fixed, with a "constant" that may depend on y:

$$F(x, y) = \int f(x, y)\, dx + C(y). \tag{10.6}$$

Then $F_x = f$ automatically. We need $F_y = g$:

$$\frac{\partial}{\partial y}\int f\,dx + C'(y) = g \quad\Longrightarrow\quad C'(y) = g(x,y) - \frac{\partial}{\partial y}\int f(x,y)\,dx. \tag{10.7}$$

This defines $C(y)$ only if the right side does not depend on x. Differentiate it in x:

$$\frac{\partial}{\partial x}\left[g - \frac{\partial}{\partial y}\int f\,dx\right] = \frac{\partial g}{\partial x} - \frac{\partial}{\partial y}\frac{\partial}{\partial x}\int f\,dx = \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} = 0$$

by (10.5). So the right side of (10.7) is a function of y alone, and integrating it gives $C(y)$.

■

EXAMPLE 10.1 (ROBINSON P. 91): $\left(x^3 + \frac{y}{x}\right) + \left(y^2 + \ln x\right)\frac{dy}{dx} = 0,\ \ x > 0$

1. $f = x^3 + y/x$, $g = y^2 + \ln x$. Test: $f_y = 1/x$, $g_x = 1/x$: exact .
2. Integrate f in x: $F = \displaystyle\int \left(x^3 + \frac{y}{x}\right)dx = \frac{x^4}{4} + y\ln x + C(y)$.
3. Match $F_y = g$: $\ln x + C'(y) = y^2 + \ln x$, so $C'(y) = y^2$ (indeed independent of x) and $C(y) = y^3/3$.
4. Solution, implicitly: $\displaystyle\frac{x^4}{4} + y\ln x + \frac{y^3}{3} = c$.

Check: differentiating with (10.2): $F_x + F_y y' = (x^3 + y/x) + (\ln x + y^2)y' = f + gy'$.

6.1 Integrating factors for non-exact equations [Robinson, Sec. 10.1.1, pp. 91–94]

Multiplying by a well-chosen $I(x,y)$ can make an equation exact. The linear equation $y' + p(x)y = 0$, written $p\,y + 1\cdot y' = 0$, is not exact ($\partial_y(py) = p \neq 0 = \partial_x 1$) but $e^{P(x)}$ makes it exact: $\partial_y\left(pe^P y\right) = pe^P = \partial_x e^P$. So the integrating factor of Section 5 is a special case.

In general, $I f + I g\,y' = 0$ is exact when $\partial_y(If) = \partial_x(Ig)$. By the product rule this is

$$I_y f + I f_y = I_x g + I g_x \quad\Longleftrightarrow\quad \left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x}\right) I = g\frac{\partial I}{\partial x} - f\frac{\partial I}{\partial y}, \tag{10.9}$$

a PDE for I that is in general no easier than the original problem. It becomes an ODE if I depends on one variable only.

SPECIAL INTEGRATING FACTORS

$I = I(x)$ (Robinson (10.10)–(10.11)): then $I_y = 0$ and (10.9) becomes

$$\frac{dI}{dx} = \frac{1}{g}\left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x}\right) I,$$

which has a solution $I(x)$ provided $\dfrac{1}{g}\left(f_y - g_x\right)$ depends on x only. Then $I(x) = \exp \displaystyle\int \frac{f_y - g_x}{g}\, dx$.

$I = I(y)$ (the symmetric case, not treated in the book): then $I_x = 0$ and (10.9) becomes

$$\frac{dI}{dy} = \frac{1}{f}\left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) I,$$

which has a solution $I(y)$ provided $\dfrac{1}{f}\left(g_x - f_y\right)$ depends on y only. Then $I(y) = \exp \displaystyle\int \frac{g_x - f_y}{f}\, dy$.

EXAMPLE 10.2 (ROBINSON PP. 93–94): AN INTEGRATING FACTOR $I(x)$

$$\left(3\sin y + 5ye^{5x} + \frac{2ye^{5x}}{x}\right) + \left(x\cos y + e^{5x}\right)\frac{dy}{dx} = 0.$$

1. $f_y = 3\cos y + 5e^{5x} + \dfrac{2e^{5x}}{x}$ and $g_x = \cos y + 5e^{5x}$ (chain rule on e^{5x}). Not exact.
2. $\dfrac{f_y - g_x}{g} = \dfrac{2\cos y + 2e^{5x}/x}{x\cos y + e^{5x}} = \dfrac{(2/x)\,(x\cos y + e^{5x})}{x\cos y + e^{5x}} = \dfrac{2}{x}$ – a function of x only.
3. $\dfrac{dI}{dx} = \dfrac{2}{x} I$, so $\ln I = 2\ln x$ and $I = x^2$.
4. New equation: $\left(3x^2\sin y + 5x^2 ye^{5x} + 2xye^{5x}\right) + \left(x^3\cos y + x^2e^{5x}\right)y' = 0$. Test: both partials equal $3x^2\cos y + 5x^2e^{5x} + 2xe^{5x}$.
5. $F = \int \tilde{g}\, dy = x^3\sin y + x^2ye^{5x} + C(x)$; then $F_x = 3x^2\sin y + 2xye^{5x} + 5x^2ye^{5x} + C'(x) = \tilde{f}$ gives $C' = 0$.
6. Solution: $x^3\sin y + x^2ye^{5x} = c$.

ILLUSTRATION – AN INTEGRATING FACTOR $I(y)$

Solve $y + \left(2x + y^2\right)\dfrac{dy}{dx} = 0, y > 0$.

1. $f = y, g = 2x + y^2$: $f_y = 1, g_x = 2$. Not exact.
2. $\dfrac{f_y - g_x}{g} = \dfrac{-1}{2x + y^2}$ depends on both variables, so no $I(x)$. But $\dfrac{g_x - f_y}{f} = \dfrac{1}{y}$ depends on y only.
3. $\dfrac{dI}{dy} = \dfrac{I}{y}$, so $I(y) = y$.

4. New equation: $y^2 + (2xy + y^3)y' = 0$. Test: $\partial_y(y^2) = 2y = \partial_x(2xy + y^3)$.
5. $F = \int y^2\,dx = xy^2 + C(y)$; $F_y = 2xy + C'(y) = 2xy + y^3$, so $C(y) = y^4/4$.
6. Solution: $xy^2 + \dfrac{y^4}{4} = c$. Check: $\frac{d}{dx}$ gives $y^2 + 2xyy' + y^3y' = y\,[\,y + (2x + y^2)y'\,] = 0$.

SECTION 07 • ROBINSON SEC. 10.2 (BEYOND THE LECTURE MAP)

7. Substitutions: homogeneous and Bernoulli equations

Robinson's Chapter 10 closes with two substitutions that turn special nonlinear equations into ones already solved. They are not in the lecture map but appear in the problem sheets' style of questions.

- **Homogeneous equations** $\dfrac{dy}{dx} = F\!\left(\dfrac{y}{x}\right)$ (a different meaning of "homogeneous" from the linear one). Put $y = ux$; by the product rule $y' = u + xu'$, so $x\dfrac{du}{dx} = F(u) - u$, which is separable. Example 10.3: $xyy' = 2x^2 + 3y^2$ gives $y = \pm x\sqrt{\alpha x^4 - 1}$.
- **Bernoulli equations** $\dfrac{dy}{dx} + p(x)y = q(x)y^n$ (10.13), $n \neq 0, 1$. Put $u = y^{1-n}$; by the chain rule $u' = (1-n)y^{-n}y' = (1-n)y^{-n}\big(-py + qy^n\big) = (1-n)(-pu + q)$, so

$$\frac{du}{dx} + (1-n)p(x)\,u = (1-n)q(x),$$

  which is linear and solved by an integrating factor.

EXAMPLE 10.4 (ROBINSON P. 96): $y' - 6xy = 2xy^2$

$n = 2$, $u = y^{-1}$, $u' = -y^{-2}y' = -y^{-2}(6xy + 2xy^2) = -6xu - 2x$, so $u' + 6xu = -2x$. Integrating factor $e^{\int 6x\,dx} = e^{3x^2}$: $\frac{d}{dx}\big(ue^{3x^2}\big) = -2xe^{3x^2}$; with $w = 3x^2$, $dw = 6x\,dx$, $\int -2xe^{3x^2}dx = -\frac{1}{3}e^{3x^2}$. Hence $u = -\frac{1}{3} + ce^{-3x^2}$ and $y = 1/u = \dfrac{3}{Ce^{-3x^2} - 1}$ with $C = 3c$.

8. Choosing a method

IF THE EQUATION IS…	RECOGNISE IT BY	DO THIS	ROBINSON
Autonomous $\dot{x} = f(x)$	No explicit t	Phase line; stability by sign of $f'(x^*)$. Solve explicitly only if rates are needed.	Ch. 7
Separable $\dot{x} = f(x)g(t)$	Right side factorises	List constant solutions $f(x^*) = 0$; then integrate $\int dx/f = \int g\,dt$; remove moduli with the phase line.	Ch. 8
Linear $\dot{x} + p(t)x = q(t)$	x and $\dot{x}$ appear to the first power	$I = e^{\int p\,dt}$, then $\frac{d}{dt}(Ix) = Iq$.	Ch. 9
Exact $f + gy' = 0$	$f_y = g_x$	Build F with $F_x = f$, $F_y = g$; answer $F = c$.	Sec. 10.1
Not exact	$(f_y - g_x)/g$ depends on x only, or $(g_x - f_y)/f$ on y only	Multiply by $I(x)$ or $I(y)$, then as above.	Sec. 10.1.1
Homogeneous / Bernoulli	$y' = F(y/x)$ / $y' + py = qy^n$	$y = ux$ / $u = y^{1-n}$.	Sec. 10.2

ALWAYS CHECK

Substitute the answer back into the equation and the initial condition. Look for lost constant solutions after dividing. Test the limits $t \to 0$ and $t \to \infty$ against the phase line. Check the dimensions of every constant that appears.